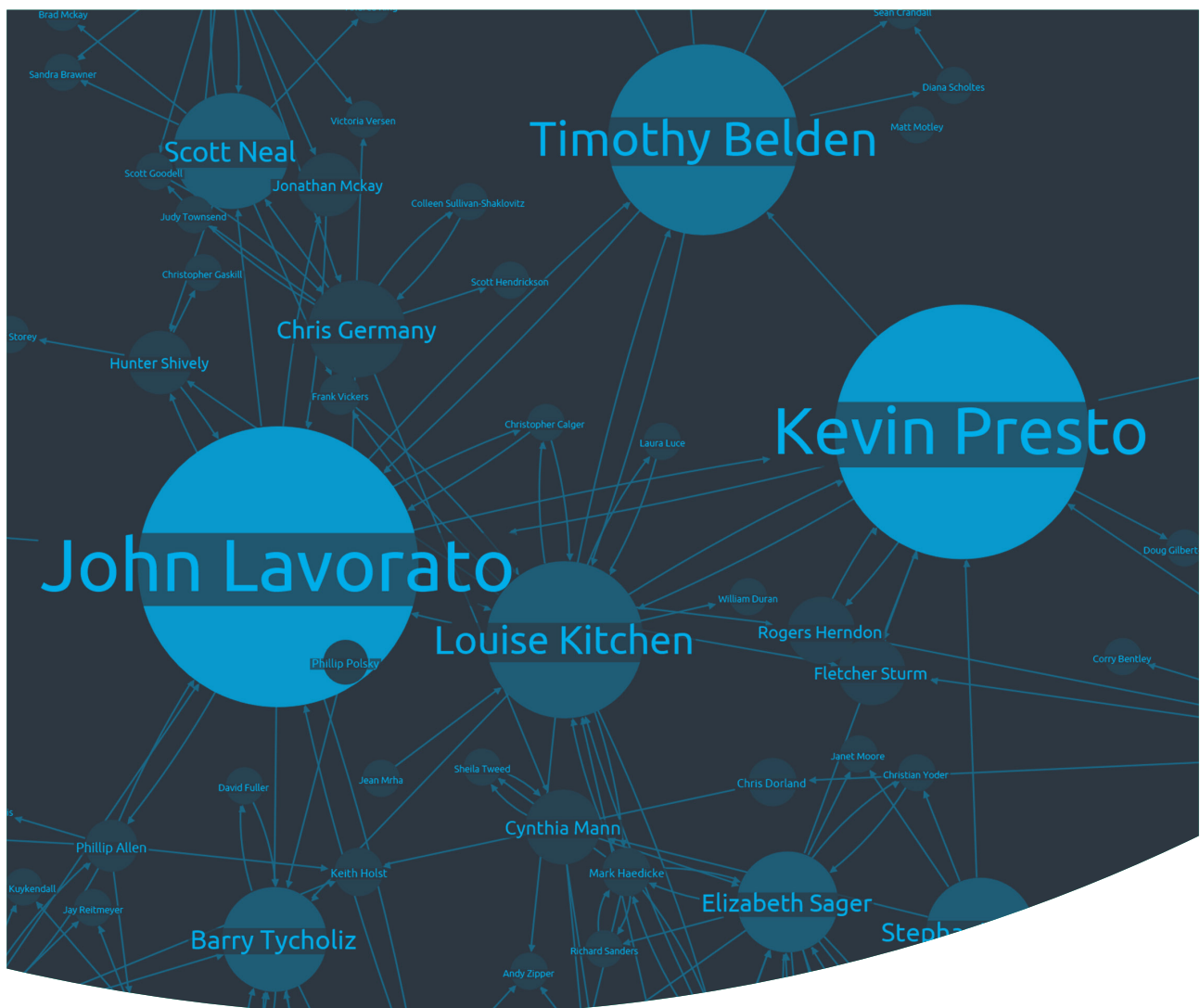
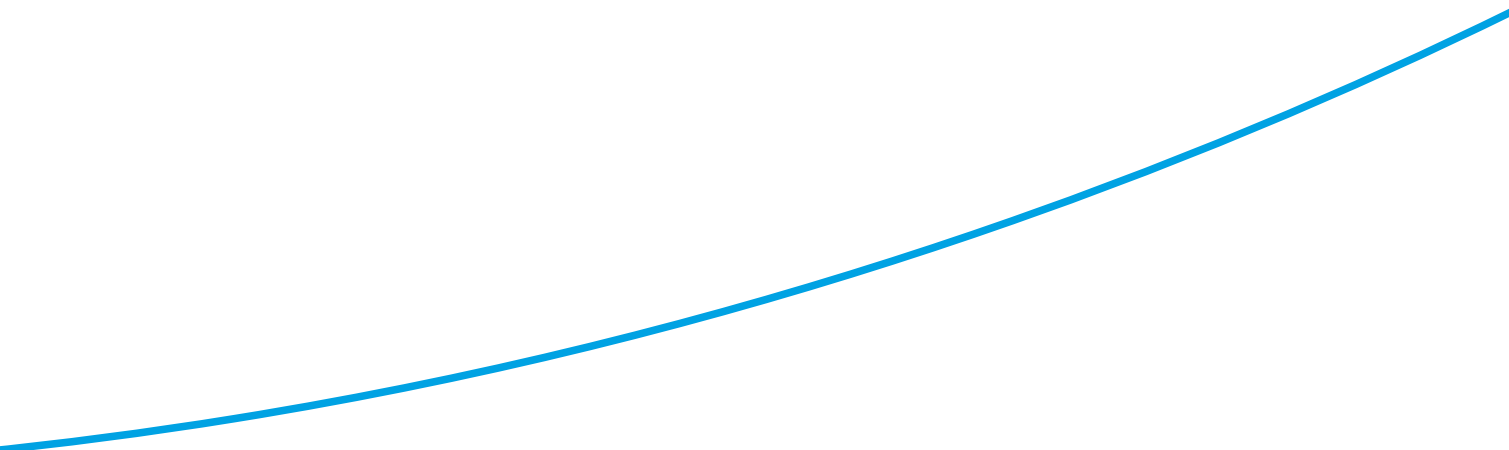


Social network analysis and visualization



Cambridge
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What is a social network?

A social network is a structure of social relationships. Visualizing these networks lets us observe behaviors, identify influence, and make predictions about how individuals or groups may act.

These relationships don't need to be between people. The methods described in this white paper apply to any kind of connected structure, from an IT network, to an organization or a supply chain. Anything that can be imagined as a set of nodes and links can be visualized and analyzed as a social network.

Why visualize a social network?

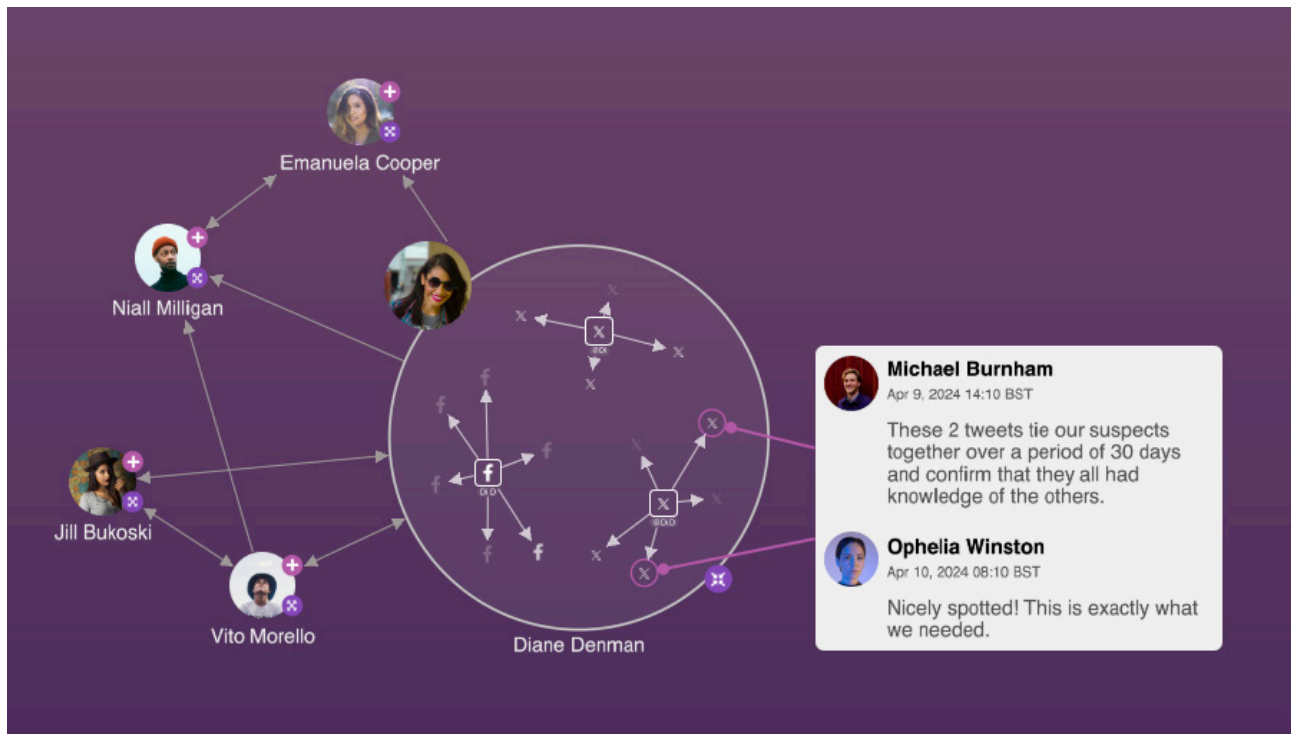
Connected data visualization, also known as network visualization, link analysis or graph visualization is so effective because it's:

Intuitive: the node-link model instantly makes sense.

Fast: our brain is great at finding trends, patterns and outliers when data is presented in a tangible format.

Flexible: the world is densely connected. As long as there's an interesting relationship in your data somewhere, you'll find value in network visualization.

Insightful: exploring connected data interactively allows users to gain deeper knowledge, understand context and ask more questions, compared to static visualization or just looking at raw data.



A screenshot of an interactive social media network visualization, with annotations from investigators

What is social network analysis?

Social network analysis (SNA) is the use of graph theory to understand and measure social networks. It explains social behavior by analyzing connections, rather than just looking at individuals in isolation. It has its roots in academic social science, but has many real-world applications, including:

Government and law enforcement

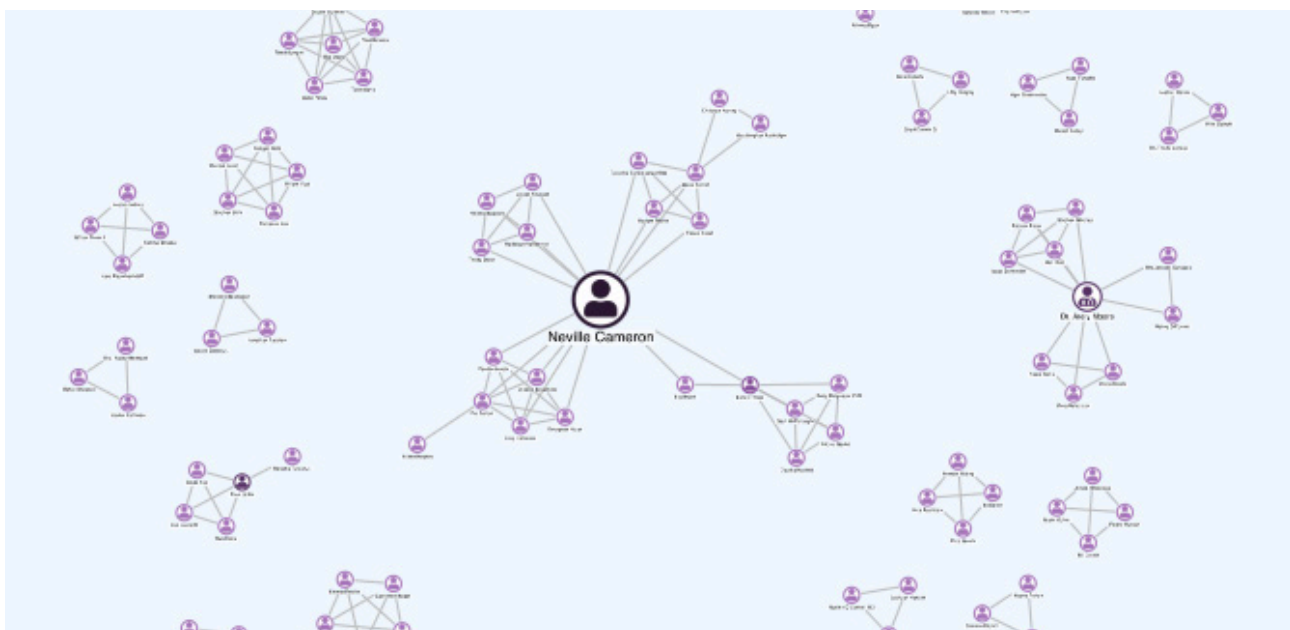
- Predict criminal activity and monitor connections between suspects
- Understand gang dynamics and structures
- Find new lines of enquiry by mapping known connections between events, times, places

Banking and finance

- Identify, find and target influential people, including decision makers and thought leaders
- Anticipate customer needs and improve service and product recommendations

Cyber security analysts and infrastructure managers

- Monitor connections between devices and events to uncover anomalies and threats
- Perform post-incident forensics



Visualizing insurance claims as a network

Breaking down a social network

We can analyze social networks at three different levels. Each of these offers different insight:

- **Network** – this high-level overview reveals patterns in a network, like overall connectivity and balance.
- **Group** – this is when a graph is viewed at a sub-network level, one segment at a time. This perspective reveals clusters of connections so it's easier to see dynamics between groups.
- **Entity** – at a node-by-node level, we can identify small clusters and understand individual entities within their connected environment.

Centrality Measures

A node's centrality is a measure of its prominence or structural importance in a network. A high centrality score could indicate power, influence, control, or status.

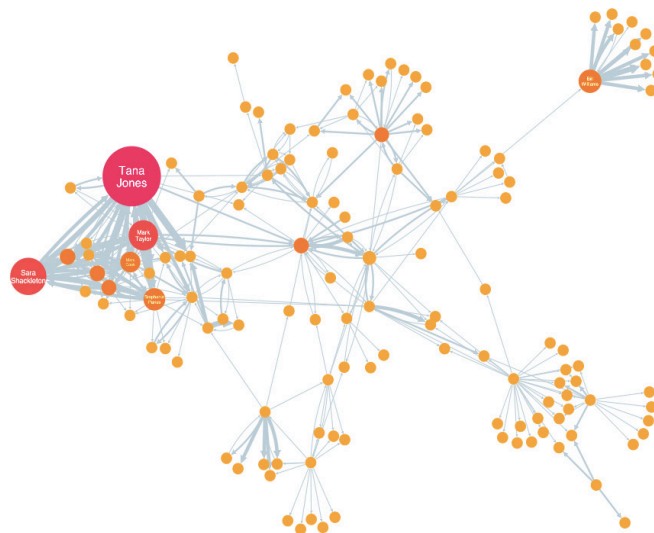
Finding out which is the most 'central' node can disseminate information in a network faster, stop epidemics, protect a network from breaking, identify suspected terrorists and much more.

Our network visualization toolkits include a range of centrality measures to identify the most influential nodes in a social network.

Degree Centrality

The degree centrality measure finds nodes with the most links to other nodes in the network. It's the simplest measure of a node's connectivity and answers questions like:

- Who is the most / least popular node in this network?
- Who can call upon the most resources in this network?



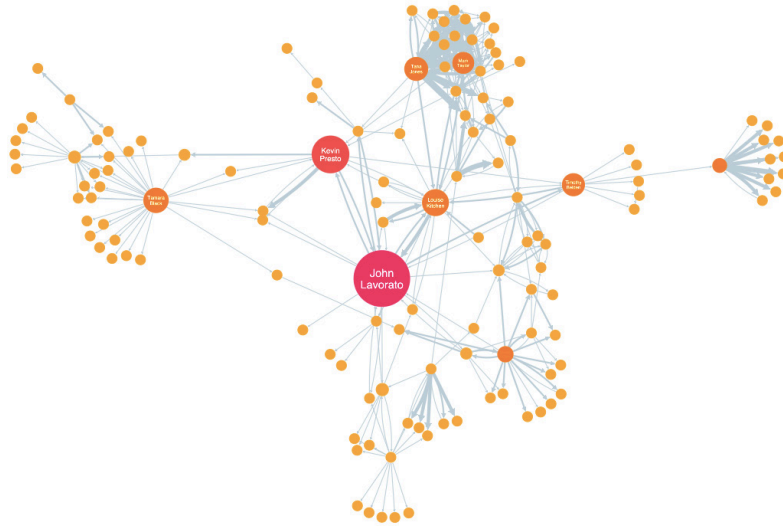
A network of nodes, sized by degree

Betweenness

Nodes with a high betweenness centrality score are the ones that most frequently act as 'bridges' between other nodes. They form the shortest pathways of communication within the network and answers questions like:

- Who or what can most strongly control information flow around the network?
- Who or what would cause the most disruption to flow if they/it were removed?

Use betweenness centrality with care: a high score could indicate an important node connecting disparate clusters, or just that a node is on the periphery of several clusters.



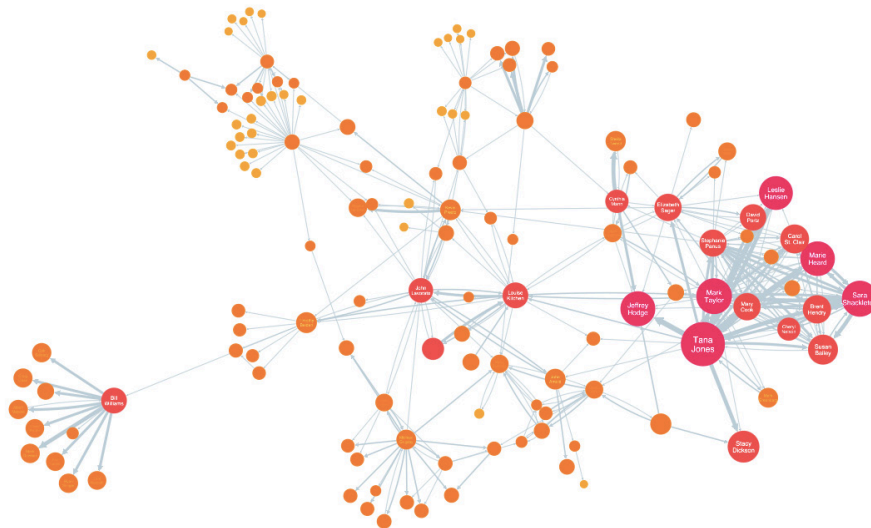
A network of nodes, sized by betweenness

Closeness

This measure is similar to betweenness but instead of calculating the number of paths through each node, it calculates a node's proximity to other nodes. To do this, it finds all of the shortest paths in a network, and then assigns each node a score based on the sum of its shortest paths.

Closeness is most insightful when a network is sparsely connected. In a highly connected network you will often find all nodes have a similar score. It answers questions like:

- Who can most efficiently obtain information on other nodes in the network?
- Who could most quickly spread information in a network?



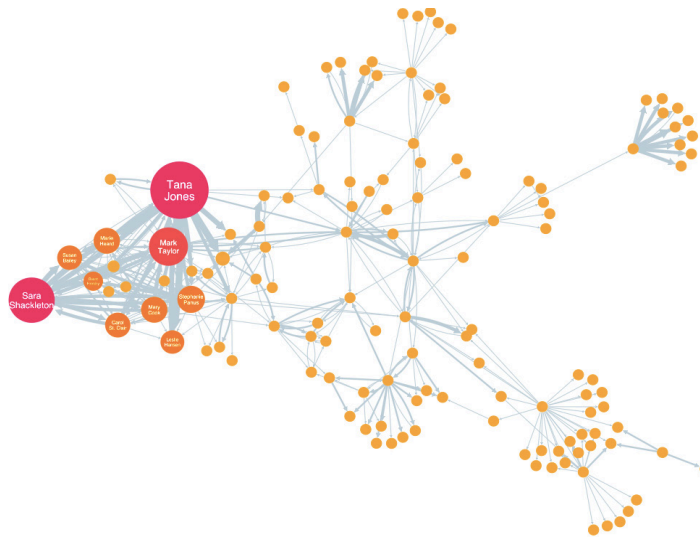
A network of nodes, sized by closeness

EigenCentrality

EigenCentrality measures the number of links each node has, the number of links their connections have, and so on throughout the network. It's useful because it indicates direct influence and influence over nodes more than one 'hop' away.

A node may have a high degree score (i.e. many connections) but a relatively low EigenCentrality score if many of those connections are with similarly low-scored nodes, and vice versa. It answers questions like:

- Who or what holds wide-reaching influence in my network?
- Who or what is important in my network on a macro scale?



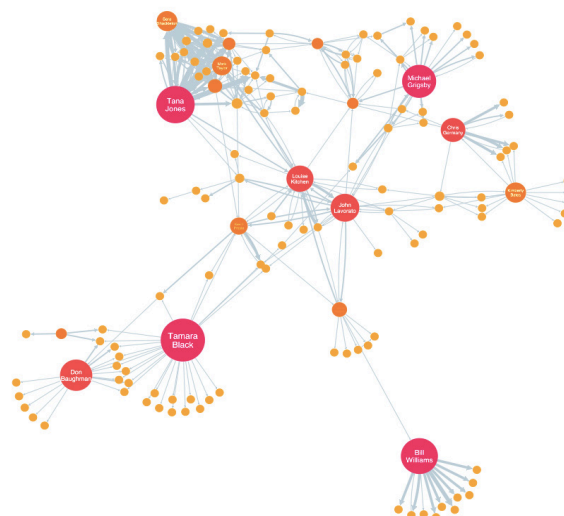
A network of nodes, sized by EigenCentrality

PageRank

Like EigenCentrality, PageRank uncovers influential or important nodes whose reach extends beyond just their direct connections. It identifies the significance of nodes by assigning a score based on the number of incoming links (its 'indegree').

PageRank is another way of answering similar questions to EigenCentrality, i.e.:

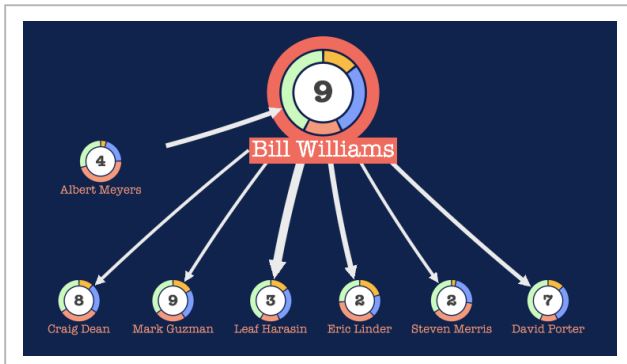
- Who or what holds wide-reaching influence in my network?
- Who or what is important in my network on a macro scale?



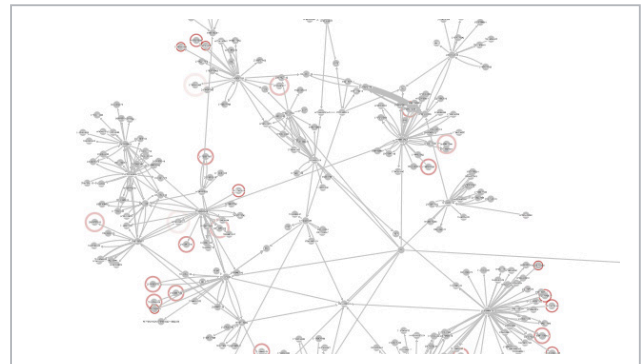
A network of nodes, sized by PageRank

How to represent SNA information

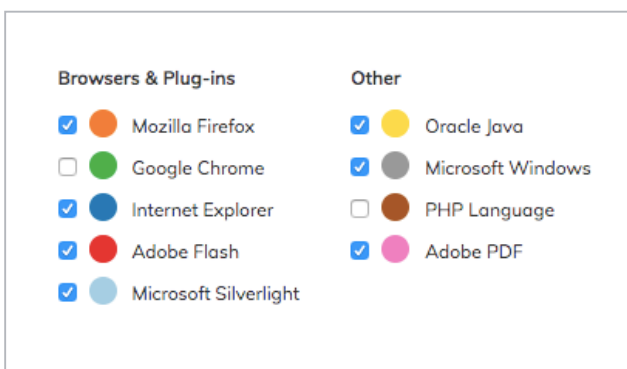
Your users need to quickly and intuitively understand SNA information in the visualization tools you build. Our toolkits make it easy for you to do this, including:



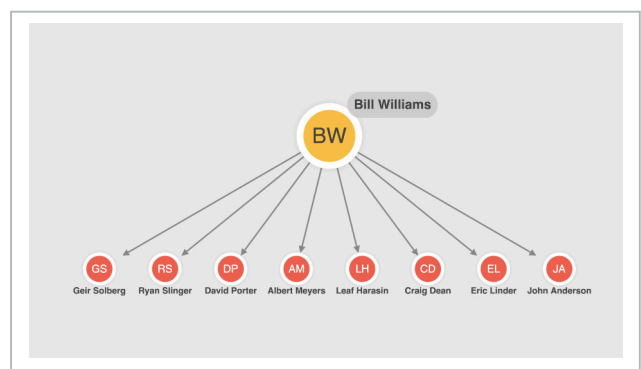
Formatting: highlight key players by coloring or re-sizing nodes, or show numeric proportions of data, such as different centrality measures, with donuts around the nodes.



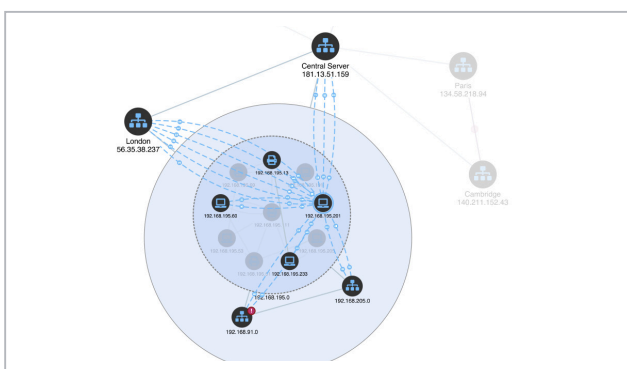
Animation: apply animation to highlight nodes of importance, or show changes of centrality.



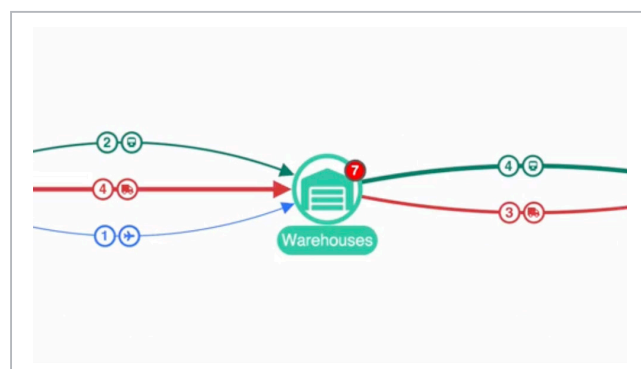
Filters: eliminate noise and focus on the connections of interest with donuts.



Glyphs, labels and annotations: add additional information with glyphs, labels, or annotations.



Combos: combine nodes together and run measures across groups of nodes, using combos.



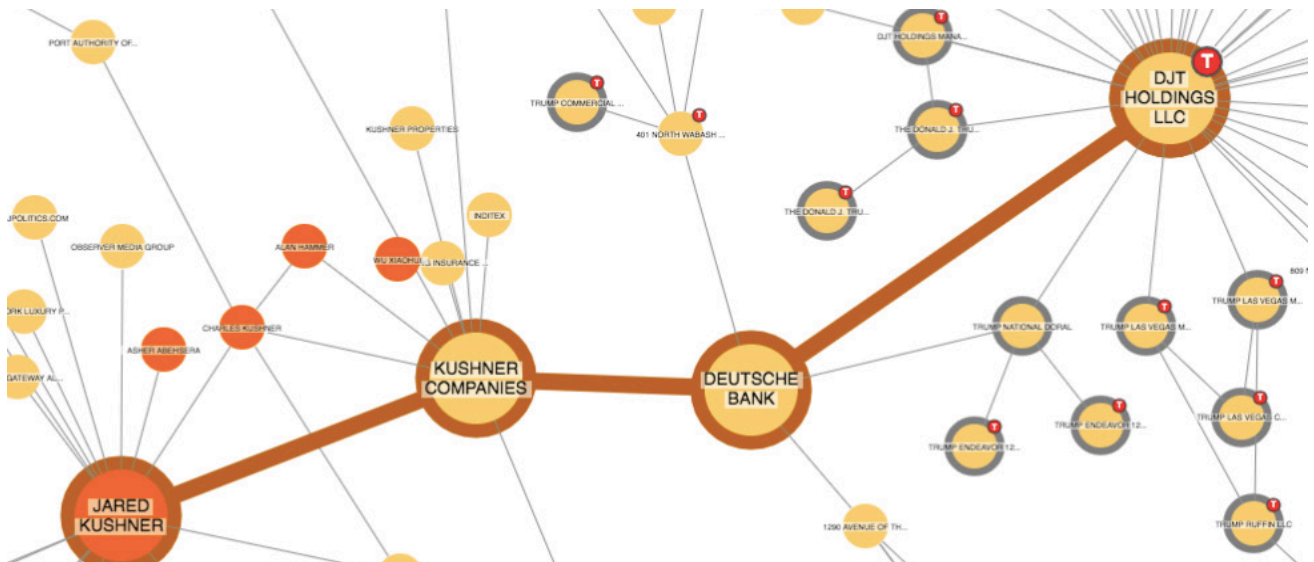
Aggregated links: multiple links clutter a chart. Aggregate them for a cleaner visualization. with the details still available on demand.

Helpful graph algorithms

Alongside centrality measures, we've developed algorithms to help your users understand social network data. These include:

Distance / shortest path

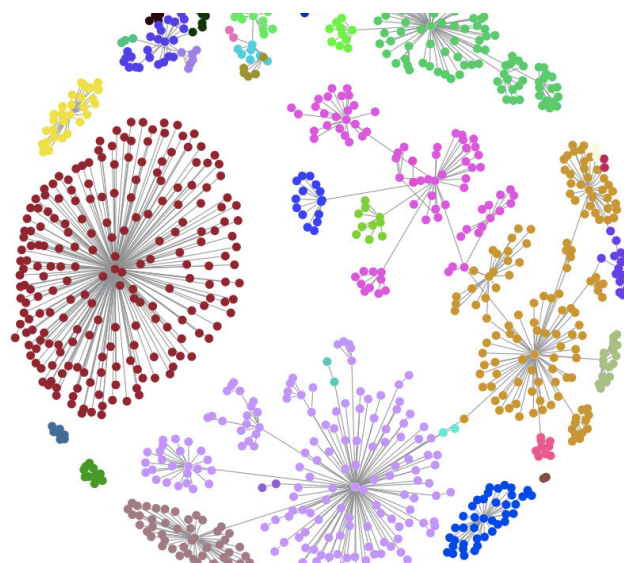
The 'distance' function measures how many hops apart two nodes are in a network. Shortest path highlights the route that passes through the lowest number of nodes. You can also add weight to hops to calculate actual distances as well as the number of hops.



The shortest path between two nodes

Community finding and clustering

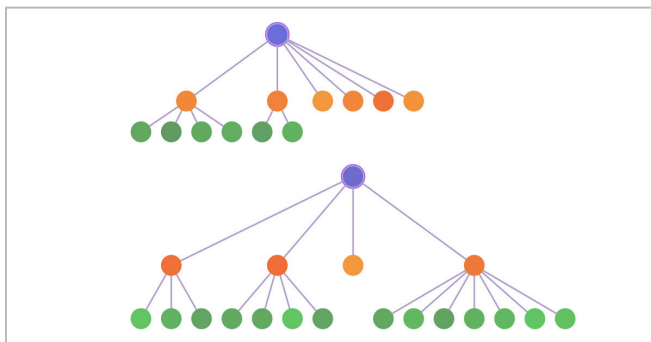
Network clustering uncover community structures in large datasets. It's based on a concept known as modularity – a way to measure how readily a network can be divided into sub-networks, or modules. These algorithms make it much easier to find communities and start learning more about the organizations being investigated.



A social network, divided into communities with a clustering algorithm

Network layouts

Layouts are algorithms that decide where nodes are placed on the screen, and are a crucial tool in social network visualization. Our layouts provide a precise and consistent way to untangle nodes and links, and present them in a meaningful way.



The standard layout is an efficient, force-directed layout, meaning it uses physics forces to ensure links are a consistent length with nodes and links distributed evenly, overlapping as little as possible.

The organic layout detangles complex networks by spreading nodes and links apart, arranging multiple components in a circular shape, with larger components in the center.

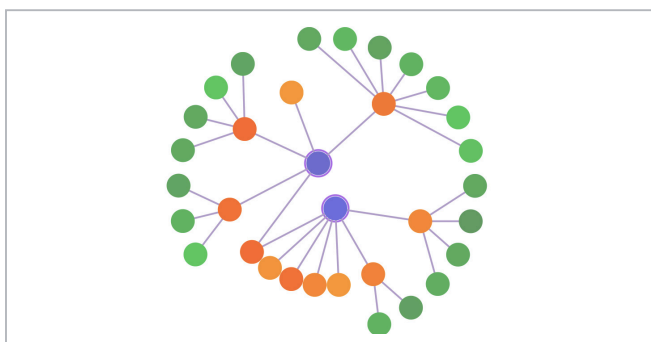
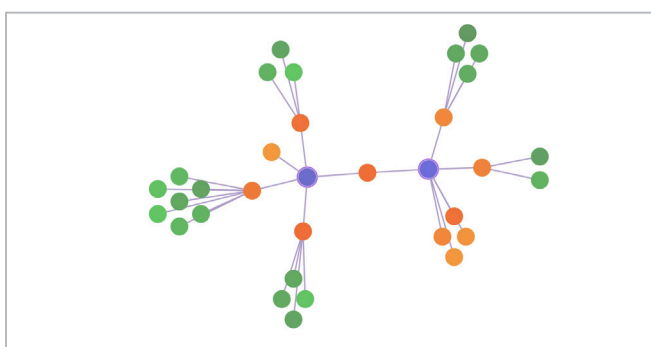
The sequential layout is a good way to display data with a clear sequence of links between distinct levels of nodes.

The hierarchy layout produces a family tree of nodes, with child nodes positioned in layers leading from their parents.

The structural layout is similar to the standard layout, but nodes with similar attributes are grouped together in fans around a central node or cluster.

The radial layout arranges nodes in concentric circles around the original subject in a radial tree. Each 'generation' of node becomes a new orbit extending outwards from the original parent as the dependency chain grows.

The lens layout pushes highly-connected nodes into the center, and force other nodes to the periphery to give an attractive 'fish-eye lens' view.



More information about these layouts, visit: cambridge-intelligence.com/layouts/

Ready to get started?

There's a huge amount of data insight buried in social networks. Our graph visualization toolkits make it quick and easy to build powerful network visualization tools to uncover it.

To start a free trial of KeyLines, our JavaScript toolkit for network visualization, or ReGraph, our React toolkit for network visualization, visit cambridge-intelligence.com/try/

Request a free trial

Real-world data is complicated. Visualizing it shouldn't be.

Our developer toolkits make it easy to build powerful, customized user experiences for complex data.

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GRAPH



KeyLines

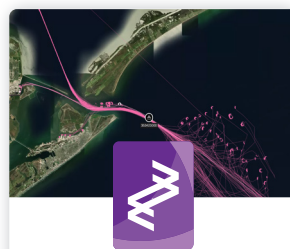
The graph visualization SDK for JavaScript developers



ReGraph

Hassle-free graph visualization for React developers

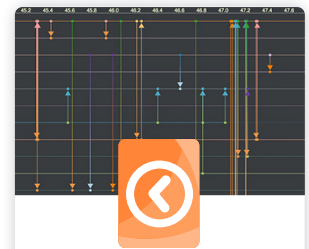
GEOSPATIAL



MapWeave

The geospatial visualization SDK that uncovers every connection

TIMELINE



KronoGraph

Advanced timeline visualizations that scale



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